



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (INFORMATION SHEET)

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TAHAP / LEVEL	3 (THREE)	
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TAJUK/TITLE: ENVIRONMENTAL CONSERVATION

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the Occupational Safety And Health (OSH) Act, Industrial Code Of Practise (ICOP) and Guidelines. The person who is competent in Health, safety and environmental consciousness shall be able to execute health consciousness, execute safety consciousness and execute environment compliance. The person who is competent in this CA shall be able to execute his/her job functions in accordance to the requirements of health, safety and environment rules and regulations.

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PENERANGAN/INFORMATION:

1. Environmental Conservation

Some people believe that public land should be managed by the method of conservation, meaning that the environment and its resources should be used by humans and managed in a responsible manner. These types of people see the value of the environment as the goods and services that it can provide to people.

This viewpoint requires that the environment be used in a way that is sustainable, and it ensures that the natural resources will be used in a manner that will meet the present day needs for the resource without jeopardizing the supply of the resource for future generations.

By using the environment sustainably, the environment and the natural resources it provides will not be depleted or destroyed permanently - and will be available for human use for a very long time. If people do not manage the land properly and the resources are not being used sustainably, then the environment can be destroyed, and the conservation method will have failed.

Environmental conservation is the act of conserving or saving our natural resources through careful management. This means we can use the resources but wisely and responsibly. Examples: reducing your waste, saving trees, recycling, using renewable resources that don't deplete our natural resources.

1.1. Conservation Concept.

Conservation is a broad concept with involves not only the scientific but ethical, moral, economic and political aspects as well. Conservation has been variously defined. Conservation for the petroleum engineer is largely minimizing of waste from incomplete extraction and for a forester it may be sustained yield of products. In all cases, conservation deals with judicious development and manner of use of natural resources of all kinds.

A generalized definition of conservation is "the maximization overtime of the net social benefits in goods and services from resources". Although it is technologically based, conservation cannot escape socially determined values.

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Conservation may also be defined as the achievement of the highest sustainable quality of living for mankind by the rational utilization of the environment, protection of nature to enrich the life of man and the control or elimination of environmental pollution in its many manifestations.

Conversation advocates practices that will perpetuate the resources of the earth on which man depends or in whose continued existence he takes an interest. Conversation derives its tenets from knowledge of ecology, the science concerned with interrelationship between living things and their environment.

But the question arises, why there is a need for conversation? The reasons are:

- i) World population is increasing at an alarming rate,
- ii) World resources are being used up at an increasing rate due to increase in population,
- iii) Pollutions is increasing with the passage of time, and
- iv) Damage caused by human activities is sometimes irreversible.

Conservation involves perpetuation of the natural environment of man including the infinite resources of air, water, soil and life forms. Conversation involves the collective responsibility of governments, private organizations, industries and individuals and the setting aside of funds, finances for ecological research and execution of conservation projects.

1.2 Aims and Principles of Conservation

The aims of conservation are two-fold:

- i) To ensure the preservation of a quality environment that considers aesthetics andrecreational as well as product needs.
- ii) To ensure a continuous yield of useful plants, animals and materials by establishing abalanced cycle of harvest and renewal.

Principles of Conservation

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Conservation is achieved through measures adopted in favor of a natural resource in order to increase its longevity and improve usage patterns. Some such measures are as follows:

i) Rational use of the resources:

Rational use of the resources is one of the concepts is conservation o natural resources in an essentially undisturbed condition because they are of scientific interest, have aesthetic appeal or have recreational value. Preservation also serves an ecological purpose by maintaining the function of the total environment, for example, protection of forests assures a sustained yield of water into urban reservoirs, and protection of estuaries perpetuates ocean fishery.

But rational use is not just preservation. It also implies the direct use of resources for their commodity or recreational value. Thus, harvesting of forest crops, livestock grazing of grassland, catching fish and hunting wild animals can be considered a legitimate part of the rational use of natural resources, if they are carried out in such a way that the resource is perpetuated and not endangered.

ii) Sustained yield:

Concept of sustained yield is involved in these activities. This means cropping the annual surplus of individuals so as not to endanger the breeding stock of game animals or fish. Similarly, tree cutting or grazing of grass should remove only the annual increment and no more.

iii) Restoration:

Restoration is another important aspect of conservation. It is a widely familiar conservation measure, which is essentially the correction of past careless activities that have impaired the productivity of the resource base. Deforests areas and mined and barren lands can be revegetated with some effort. Depleted animal and plant populations can recover if they are accorded protection. This measure is familiar in modern soil and water conservation practices applied to agricultural land.

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Restoration is possible, however, only as long as species are protected and genetic diversity of life is maintained. When species become extinct, the restoration of past conditions becomes impossible.

iv) Protection:

Protection of natural resources from commercial exploitation to prolong their use for recreation, watershed protection, and scientific study. This is the concept underlying the establishment and protection of parks and reserves of many kinds.

v) Reutilisation :

Reutilisation is the reuse of waste materials, as in the use of industrial water after it has been purified and cooled. The same process becomes recycling if the water material requires minor treatment before it can be used, as in the use of scrap iron in steel manufacture.

vi) Substitution:

Substitution, an important conservation measure, has two aspects: (i) the use of a common resource instead of a rare one when it is for the same purpose, (ii) the use of a renewable rather than a non-renewable resource when conditions permit.

vii) Allocation:

Allocation concerns the strategy of use- the best use of a resource. For many resources and their products, the market price decides as to the use a resource is put, but under certain instances, general welfare may dictate otherwise. The allocation of resources may be controlled by government through the use of quotas, rationing and outright permits.

viii) Integration:

Integration in resource management is a conservation measure because it maximizes over a period of time, the sum of goods and services that can be had

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from a resource, or a resource complex such as river valley. This is preferable to maximize certain benefits from a single resource at the expense of other benefits or other resources. Integration is a central objective of planning.

2. National Policy On The Environment

The National Policy on the Environment which integrates the three elements of sustainable development: economic, social and cultural development and environmental conservation was formulated and approved in 2002. The Policy aims at continued economic, social and cultural progress and enhancement of the quality of life of Malaysians through environmentally sound and sustainable development. It is based in eight (8) inter-related and mutually supporting principles set to harmonize economic development goals with environmental imperatives:-

- Stewardship of the Environment
- Conservation of the Nature"s Vitality and Diversity
- Continuous Improvement in the Quality of the Environment
- Sustainable Use of Natural Resources
- Integrated Decision-making
- Role of the Private Sector
- Commitment and Accountability
- Active Participation in the International Community

In keeping abreast with the country"s rapid economic development and to meet with the nation"s aspiration for an improved quality of life, the National Policy on the Environment serves as an important guide to all stakeholders to ensure that the environment is clean, safe, healthy and productive.

2.1 Environmental Quality Act, 1974

The legislation that is related to the prevention, abatement, control of pollution and enhancement of the environment in Malaysia is the Environmental Quality Act, 1974. The Act restricts the discharge of wastes into the environment in contravention of the acceptable conditions. To date 38 sets of Regulations and Orders as per *Appendix A* have been introduced and enforced. The Director General of Environmental Quality

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has been appointed by the Minister to administer this Act and any regulations and orders made thereunder.

2.2 Environmental Requirements

Under the Environmental Quality Act (EQA), 1974 and the Regulations thereunder, industrial activities are required to obtain the following approvals from the Director General of Environmental Quality prior to project implementation:

- i) Environmental Impact Assessment reports - under Section 34A of the EQA, 1974 (for prescribed activities);
- ii) Site suitability evaluation (for non-prescribed activities);
- iii) Written permission to construct - under Section 19 of the EQA, 1974 (for prescribed premises-scheduled wastes treatment and disposal facilities, crude palm oil mills and raw natural rubber processing mills);
- iv) License to use and occupy prescribed premises and prescribed conveyances - under Section 18 of the EQA, 1974.

2.3 Environmental Impact Assessment (EIA) For Prescribed Activities

i) Prescribed Activities

All prescribed activities need to obtain EIA approval from the Director General of Environment prior to the giving of approval by the relevant Federal or State Government authority for the implementation of the project. The Approving Authority is the Government Authority that has the task of deciding, whether or not a project should proceed.

Every industrial proposal should be examined by the investor to see whether an environmental impact assessment (EIA) needs to be conducted. A prospective investor should therefore first of all determine whether or not a proposed venture is categorized as prescribed activity as stipulated in the Environmental Quality (Prescribed Activities) (Environmental Impact Assessment) Order 1987, (**Appendix B**).

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If the proposed venture is categorised as a “prescribed activity” under the Environmental Quality (Prescribed Activities) (Environmental Impact Assessment) Order 1987 (**Appendix B**), an EIA study needs to be conducted and the EIA report has to be submitted to the Director General of Environmental Quality for approval. The project is not allowed to proceed unless approval of the EIA report has been granted.

Due to the sensitivity of the project and polluting potential (significant impacts) from the construction and/or operations, some of the prescribed activities have been required to go through the Detailed EIA Procedures which involves public participation. Those activities are as in **Appendix C**.

ii) EIA Study and Report

An EIA Study has to be conducted by competent individuals who are registered with the Department of Environment under the EIA Consultant Registration Scheme. The DOE will reject EIA reports which are conducted by individuals who are not registered with the Department. As such, the project proponent or EIA study team leader has to ensure that all members in the EIA study team are registered with the Department of Environment. The list of registered EIA Consultants (including Subject Consultants and Assistant Consultants) is available at the Department of Environment website (www.doe.gov.my). Details on the registration scheme are also available at the website (EIA Consultant Registration Guidance Document).

The preparation of EIA reports by registered EIA consultants shall be in accordance with the guidelines prescribed by the Director General of Environment and other relevant guidelines published by other agencies. “A Handbook of Environmental Impact Assessment Guidelines” (fourth edition) 2007 has been prepared to assist project proponents understand the objectives of EIA, procedures for carrying out EIA studies and guidelines on preparation of EIA reports. Subsequently EIA Guidelines for specific activities were developed and published by the Department of Environment (list in **Appendix D**). The handbook can be obtained from the headquarters of the Department of Environment, Putrajaya, the State Offices of the Department of Environment and the office of the representative of the Department

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of Environment at the Malaysian Industrial Development Authority (MIDA). Addresses and telephone numbers of these offices are as per **Appendix E**.

iii) EIA Procedure

There are two EIA procedures adopted in Malaysia, namely the Preliminary EIA and the Detailed EIA, that can be described as follows:

(a) Preliminary EIA

Preliminary EIA is assessment of impacts due to those activities that are prescribed. The Preliminary EIA report is reviewed by a Technical Committee consisting of the Department of Environment State Offices and other relevant government agencies.

The procedure for Preliminary EIA is as shown in **Appendix F1**. The number of Preliminary EIA report to be submitted to the Department of Environment State Offices for review is 12 copies, and 3 copies plus 2 softcopies of the Executive Summary of the Preliminary EIA report, to the Department of Environment Headquarters and relevant State Office.

(b) Detailed EIA

Detailed EIA is a procedure undertaken for those projects with major/significant impacts to the environment. The procedure for Detailed EIA is as shown in **Appendix F2**. The Detailed EIA involves EIA report display for the public and affected community to comment. Activities which need to go through the Detailed EIA procedure, are listed in **Appendix C**. Notwithstanding the list in **Appendix C**, the Director General of Environment may request a Detailed EIA for other prescribed activities as he deems necessary. For projects which have been determined to require a Detailed EIA, the project initiator must submit the terms of reference (TOR) for DOE's approval, in accordance to the format outlined in specific EIA guidelines. The number of copies of TOR to be submitted to the Department of Environment

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is 35 copies. The number of Detailed EIA report to be submitted to the Department of Environment Headquarters for review is 50 copies.

For an industrial project, the EIA generally would assist in determining site suitability as well as the necessary environmental control and mitigation measures. The objectives of EIA are summarised as follows:

- To examine and select the best from the project options available;
- To identify, predict and assess significant residual environmental impacts;
- To recommend and incorporate into the project plan, appropriate abatement and mitigating measures; and
- To identify the environmental costs and benefits of the project to the community.

2.4 Site Suitability Evaluation For Non-Prescribed Activities

One of the most important procedures which have an immediate bearing on the period of processing and condition of approval on environmental ground is the SITE SUITABILITY for the proposed project. Irrespective of whether the proposed industrial activity is going to be sited within an industrial estate or otherwise, it should be developed and managed with environmentally sound control measures. Therefore, all potential industrial sites for the establishment of new industrial activities which are NOT subject to EIA Order, 1987, particularly the Small and Medium Scale Industries (SMIs), are advised to refer to the Department of Environment for consideration and advice on site suitability.

In considering the suitability, the site is evaluated in terms of its compatibility with respect to the gazetted structure/local plans, surrounding landuse, provision of set-backs or buffer zones, the capacity of the area to receive additional pollution load, and waste disposal requirements. Details on the appropriate buffer zone with respect to a specific category of industry can be obtained from "Guidelines for the Siting and Zoning of Industries" prepared by DOE. An outline of the guidelines is given in **Appendix G**.

For potentially hazardous* type of industries, the project proponent may be required to submit a RISK ASSESSMENT study to the Department of Environment as part of the site consideration, in accordance with the *EIA Guidelines For Risk Assessment 2004*, established by the DOE. [Note: *Hazardous Industry: Any industry or installation which

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has the potential for causing injury threat to health, death and damage to property or the environment].

The Term of Reference for the preparation of an EIA report for proposed industrial projects located within gazetted and EIA approved industrial estates is as in **Appendix H**. The scopes to be studied in the EIA report are not restricted to the information in this Appendix. The EIA report has to be more specific in terms of the proposed site and project.

2.5 Licence To Occupy Prescribed Premises And Prescribed Conveyances

A licence is required to occupy and operate prescribed premises namely as below:

- i) Crude palm oil mills,
- ii) Raw natural rubber processing mills, and
- iii) treatment and disposal facilities of scheduled wastes

Application shall be made after obtaining written permission and written approval. Licensing fee will be charged for every licence issued for palm oil, raw natural rubber processing mills and facilities for treatment and disposal of schedule waste, and prescribed conveyances.

Starting from 15 August 2005, licence is required to use prescribed conveyances as stipulated in the Environmental Quality (Prescribed Conveyance) (Scheduled Wastes) Order 2005. Conveyance which is categorized as prescribed conveyance namely: any vehicle or ship of any description which is:-

- i) propelled by a mechanism contained within itself;
- ii) constructed or adapted to be used on land or water; and
- iii) used for the movement, transfer, placement or deposit of scheduled wastes.

Applications for the licence shall be made after obtaining written permission and/ or written approval.

2.6 Notification For A New Source Of Sewage, Industrial Effluents And Leachate Discharge Or Release

Starting from October 2009:-

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- i) no person shall, without prior written notification to the Director General, discharge or release or permit the discharge or release of **sewage** onto or into any soil, or any inland waters or Malaysian waters. The written notification to the Director General shall be in the form as specified in the First Schedule of the Environmental Quality (Sewage) Regulations, 2009 (P.U.(A) 432/2009).
- ii) no person shall, without prior written notification to the Director General, carry out any work on any solid waste transfer station or landfill, or construct on any land any facility or building that may result in a new source of leachate discharge or release. The written notification to the Director General shall be accompanied by the information as as specified in the First Schedule of the Environmental Quality (Control of Pollution From Solid Waste Transfer Station And Landfill) Regulations, 2009 (P.U.(A) 433/2009).
- iii) no person shall, without prior written notification to the Director General-
 - a) carry out any work on any premises that may result in a new source of discharge of industrial effluent or mixed effluent;
 - b) construct on any land, building or facility designed or used for a purpose that may cause the land or building or facility to result in a new source of discharge of industrial effluent or mixed effluent;
 - c) make or cause or permit to be made any change of, to, or in any plant, machine, or equipment used or installed at the premises that causes a material change in the quantity or quality of the discharge or release from an existing source; or
 - d) carry out upgrading work of an existing industrial effluent treatment system that may result in a material change in the quantity and quality of the discharge or release.

The written notification to carry out any work, construction, or upgrading, or to make any change shall be submitted to the Director General in the form as specified in the Second Schedule of the (Industrial Effluent) Regulations, 2009 (P.U.(A) 434/2009).

2.7 Gaseous Emission

For industries emitting gaseous and air emission, they are required to comply with the following air emission standards for the control of air pollution and gaseous emissions:-

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- i) Emission Standards from Environmental Quality (Clean Air) Regulations 2014
(Appendix J1- Standard must update release CA 2014)
- ii) Recommended Malaysian Air Quality Guidelines (Ambient Standards)
(Appendix J2)

All industrial projects subject to EIA shall be designed and operated using Best Available Techniques (BAT) in achieving a high and acceptable level of protection for the environment.

2.8 Sewage, Industrial Effluent And Leachate Discharge

Industries discharging sewage, industrial effluent and leachate are required to comply with the following relevant discharge limits as stipulated in their respective regulations:-

- i) Sewage discharge standards. (Appendix K1)
- ii) Industrial effluent discharge limits. (Appendix K2)
- iii) Leachate discharge standards (Appendix K3)

2.9 Control Use Of Ozone Depleting Substances (ODS)

ODS are categorised as environmental hazardous substance under the Environmental Quality Act, 1974 (Amendment) 1996. These substances are listed as per **Appendix L**. New investments relating to the use of these substances are prohibited. Existing industries are encouraged to develop and use substitutes and to change their ODS dependent processes as soon as possible.

2.10 Scheduled Wastes Management

Malaysia has developed a comprehensive set of legal provisions related to the management of toxic and hazardous wastes. The regulation was based on the cradle to grave principle where generation, storage, transportation, treatment and disposal are regulated. A facility which generates, stores, transports, treats or disposes scheduled waste is subject to the following regulations:

- i) Environmental Quality (Scheduled Wastes) Regulations 2005;
- ii) Environmental Quality (Prescribed Conveyance) (Scheduled Wastes) Order 2005;
- iii) Environmental Quality (Prescribed Premises) (Scheduled Wastes Treatment and Disposal Facilities) Order 1989;

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- iv) Environmental Quality (Prescribed Premises) (Scheduled Waste Treatment and Disposal Facilities) Regulations 1989;
- v) Customs (Prohibition of Export) Order (Amendment)(No. 2) 1993, and;
- vi) Customs (Prohibition of Import) Order (Amendment)(No. 2) 1993.

A summary of the environmental requirements on scheduled wastes is given in *Appendix M* and the list of scheduled wastes is as per *Appendix N*.

2.11 Incentives For The Storage, Treatment And Disposal Or Toxic And Hazardous Wastes

To encourage proper industrial waste management, the following incentives are currently available:

- i) Pioneer Status incentive for 5 years to companies which are principally engaged in an integrated operation for the storage, treatment and disposal of toxic and hazardous wastes;
- ii) As a further incentive for both the above categories of companies, the Government also extends the current import duty and sales tax exemption scheme for machinery, equipment, raw materials and components.

All facilities for storage, treatment and disposal of toxic and hazardous wastes must be approved by the Department of Environment before the application is made for the incentives.

2.12 Incentives For The Installation Of Pollution Control Equipment

Under Income Tax Act 1967, Income Tax (Qualifying Plant Allowances) (Control Equipment) Rules 1998, the Government has provided special capital allowance incentive for the Companies which install pollution control equipment in the setting up of the plants. This allowance is at initial rate 40% and an annual rate of 20% for the qualifying plants stipulate under Schedule 3 of Income Tax Act 1997.

3 Environmental Hazard Control

3.1 Environmental Hazards

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i) Air

Air pollution may be found in large cities throughout the world; its sources are often attributed to automobile exhaust and industrial emissions and may be aggravated by climate and geography. Specifically, particulate matter (PM), or particle pollution, consisting of fine particles 2.5 µm or smaller in diameter, may enter the lungs and cause serious health problems. Travelers should be aware that global long-term average PM_{2.5} concentrations have been estimated to exceed the World Health Organization's Air Quality PM_{2.5} Interim Target-1 (35 µg/m³ annual average) in eastern and central Asia and North Africa.

Although the harmful effects of air pollution are difficult to avoid when visiting some cities, limiting strenuous activity and not smoking can help. Any risk to healthy short-term travelers to such areas is probably small, but people with preexisting health conditions (such as asthma, chronic obstructive pulmonary disease, or heart disease) could be more susceptible. Avoiding dust clouds and areas of heavy dust or haze is wise.

ii) Water

Rivers, lakes, and oceans may be contaminated with organic or inorganic chemical compounds (such as heavy metals or other toxins); harmful algal blooms (cyanobacteria) that can be toxic both to fish and to people who eat the fish, or who swim or bathe in the water; and pathogens from human and animal waste that may cause disease in swimmers. Such hazards may not be immediately apparent in a body of water. Available drinking water may also be contaminated;

Extensive water damage after major hurricanes and floods increases the likelihood of mold contamination in buildings. Travelers may visit flooded areas overseas as part of emergency, medical, or humanitarian missions. Mold is a more serious hazard for people with conditions such as impaired host defenses or mold allergies. To prevent exposure that could result in adverse health effects from disturbed mold, people should adhere to the following recommendations:

- a) Avoid areas where mold contamination is obvious.

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- b) Use personal protective equipment (PPE), such as gloves, goggles, and a tight-fitting approved N-95 respirator. Travelers should take sufficient PPE with them, as these may be scarce in the countries visited.
- c) Keep hands, skin, and clothing clean and free from mold-contaminated dust.
- d) Review the CDC guidance, Mold Prevention Strategies and Possible Health Effects in the Aftermath of Hurricanes and Major Floods which provides recommendations for dealing with mold in these settings.

iii) Radiation

Natural background radiation levels can vary substantially from region to region, but these natural variations are not a health concern for either the traveler or resident population. Travelers should be aware of regions known to have been contaminated with radioactive materials, such as the areas surrounding the Chernobyl nuclear power plant in Ukraine and the Fukushima Daiichi nuclear power plant in Japan.

The Chernobyl plant is located 100 km (62 miles) northwest of Kiev. This 1986 accident contaminated regions in 3 republics—Ukraine, Belarus, and Russia—with the highest radioactive ground contamination within 30 km (19 miles) of Chernobyl.

The Fukushima Daiichi plant is located 240 km (150 miles) north of Tokyo. The area within a 20-km (32-mile) radius of the plant is restricted, and Japanese authorities also advised evacuation from locations farther away to the northwest of the plant. This incident occurred in 2011, and as Japanese authorities continue to clean the affected areas and monitor the situation, travel advisories may change. US travelers are advised to check the website of the US embassy in Tokyo for up-to-date information. There are no travel advisories for Tokyo or any city or region south of Tokyo. Travelers who choose to reside for >1 year within 80 km of the Fukushima Daiichi nuclear plant should consult with local authorities to receive guidance on expected levels of radiation and recommendations for reducing exposure to radiation. In addition, pregnant women, children, and the elderly should avoid residing within 30 km of the Fukushima Daiichi Nuclear Plant.

More than 450,000 travelers to the United States originate from Japan each month. During the height of the Fukushima releases, there was some concern about those

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travelers bringing contamination into the United States with them. Based on radiologic contamination screening at points of entry into the United States, 3 travelers arriving from Japan after the incident had low levels of contamination and were not considered to pose a health hazard to themselves or others.

In most countries, known areas of radioactive contamination are fenced or marked with signs. These areas should not be trespassed. Any traveler seeking long-term (more than a few months) residence near a known or suspected contaminated area should consult with staff of the nearest US embassy and inquire about any applicable advisories in that area regarding drinking water quality or purchase of meat, fruit, and vegetables from local farmers. Radiation emergencies are rare events. In case of such an emergency, however, travelers should follow instructions provided by local emergency and public health authorities. If such information is not forthcoming, US travelers should immediately seek advice from the nearest US embassy.

Natural disasters (such as floods) may also displace industrial or clinical radioactive sources. In all circumstances, travelers should exercise caution when they encounter unknown objects or equipment, especially if they bear the radioactive symbol. Travelers who encounter a questionable object should notify authorities.

3.2 Control Of Specific Environmental Hazards

Water and food Quality standards are set for the physical, chemical and biological characteristics of drinking water and monitored throughout the water treatment and distribution network through the household tap. Among the battery of tests and specified limits are those relating to pH, microorganisms including marker agents such as faecal coliform bacteria (E coli), Cryptosporidium and Giardia lamblia, chlorination by-products, dissolved metals, salts and metalloids (lead, mercury, arsenic, etc.), organic matter and radon.

Biological and chemical testing is also carried out in the food industry, where, as with water safety, the approach is typically based on the principles of the Hazard Analysis and Critical Control Points (HACCP), which seeks to identify potential hazard points, so that key actions, known as Critical Control Points (CCP's) can be taken to reduce or

eliminate the risk of the hazards being realized. The system is used at all stages of food production and preparation processes including packaging, distribution, etc.

The air quality monitoring network has 51 monitoring station of which 44 are designed to measure Carbon Dioxide (CO₂), Sulphur Dioxide (SO₂), Nitrogen Oxides (NO_x), Particulate Matter (PM₁₀) and Ozone (O₃) and 7 are designed to measure PM₁₀ only. In addition, there are 25 station for measuring total suspended particulates (TSP), PM₁₀ and heavy metal. The air quality status for Malaysia is determined and disseminate to the Air Pollution Index (API). To determine API for a given time period, the sub-index values (sub-API) for five air pollutants included in the API system are first calculated for the air quality data collected from the Continuous Air Quality Monitoring stations. The corresponding air quality data are subjected to the necessary quality control processes and quality assurance procedures, prior to the sub-index calculations. The API value reported for a given time period represents the highest API value among all the sub-APIs calculated during that particular time period. Table below shows the API values and the level of pollution and the health measure.

API	Status	Level of Pollution
0–50	Good Low	no ill effects on health
51–100	Moderate	Moderate pollution, no ill effects on health
201–300	Very unhealthy	Significant aggravation of symptoms and decreased exercise tolerance in persons with heart or lung disease
301–500	Hazardous	Severe aggravation of symptoms and a danger to health
Above 500	Emergency	Severe aggravation of symptoms and a danger to health

The government authorities are required to review and assess the current, and likely future, air quality objectives. Measures and action plan implemented to achieve clean air may include:

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- work with relevant authorities responsible for highways and/or environmental regulation on possible emissions reduction measures;
- local traffic management schemes, including use of low emission zones and congestion charging in some areas where appropriate;
- commitment to developing or promoting green travel plans and/or to using cleaner-fuelled vehicles;
- information dissemination to the public;
- partnerships with bus or fleet operators to deliver cleaner, quieter vehicles in return for better bus lanes or more flexible delivery arrangements.

Noise is a complex exposure, as its effects on health and well-being are function of multiple parameters, including: its intensity, its duration, its intermittency, its nature and quality, and its origin. Loud noise may cause deafness, but constant background noise (e.g. from heavy traffic) is often tolerated better than intermittent noise from a neighbour. The nuisance value of noise is more than just the level of sound, and the sources of noise that is most often the source of complaints is neighbours. The Department of the Environment, Food and Rural Affairs (DEFRA) have begun to generate noise maps to provide an overview of the ambient noise climate in cities and major transportation sources in England. The mapping was carried out in line with the Department's work to implement the EU's Environmental Noise Directive.

Ionizing and electro-magnetic radiation Monitoring and protection against radiation risks covers a number of important areas:

- The government recommends that households in high radon areas have radon monitoring carried out for a test period of three months. There are various engineering solutions available to reduce levels if found to be above the action limit of 200 Bq.m⁻³.
- Workers in the nuclear industry, or in other fields where there is exposure to ionizing radiation (e.g. medical imaging), are required to wear radiation badges to monitor personal exposure. Removal from further exposure is required if a personal dose limit is reached.

For non-ionizing radiation sources, there are published guidelines on exposure limits to static magnetic fields and to time-varying electric, magnetic and electromagnetic fields (up to 300 GHz). The guidelines advise basic restrictions to provide protection against the established adverse health effects of exposure.

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A disease cluster implies any unusual aggregation of cases of illness or disease that are grouped in space and time. The concept is somewhat complicated, however, because it begs the question of what is unusual.

The issue often arises that someone notices an apparently high number of cases of disease in a particular street or locality, and wonders whether there might be a connection with a local environmental problem. But in such circumstances it is very difficult to interpret the significance of the observation, as the cases may reflect no more than normal variation in time and space. The difficulty lies in the fact that the hypothesis of an unusual level of disease occurrence is formulated after seeing that there appears to be a high number of cases.

A formal statistical test might indicate a higher than average number of cases locally but that should not be surprising as the only reason one is looking is because someone has noticed that there does appear to be more cases than usual. But it may be no more than a chance occurrence. The difficulty is that the hypothesis of a 'cluster' is post hoc – developed after the observation – and therefore almost impossible to interpret appropriately.

Because of this most epidemiologists realise that there is usually very limited scientific and public health value in investigating the majority of observations of disease clusters. These arguments were very well articulated by Rothman (1990), who concluded that:

- i) With very few exceptions, there is little scientific or public health purpose to investigate individual disease clusters at all;
- ii) There is likewise very little reason to study overall patterns of disease clustering in space-time; and
- iii) As a consequent of points 1 and 2, no statistical methodologies are needed to refine our study of disease clusters or clustering in general.

And indeed, in practice it has been found that very few cluster investigations ever reveal an apparent underlying environmental cause. Most such investigations are inconclusive.

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However, it is recognized that apparent disease clusters do have to be taken seriously, if only to allay public concerns, and various proposals have been made about staged investigations that only lead to formal epidemiological studies when the case appears very strong.

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SOALAN/QUESTION:

1. Explain the environment conservation concept.

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2. What activities need through Environmental Impact Assessment (EIA) ?

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3. How Malaysian government control the management of toxic and hazardous wastes?

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RUJUKAN / REFERENCES:

1. Environmental Quality Act 1974 (Act 127): regulations, rules & orders, (2009) ISBN 978-967891-934-0
2. (<http://wwwnc.cdc.gov/travel/yellowbook/2014/chapter-2-the-pre-travel-consultation/water-disinfection-for-travelers>)